Trainings

Preparatory Steps

WARNING

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

WARNING

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/10/10-008-018-tr.html)

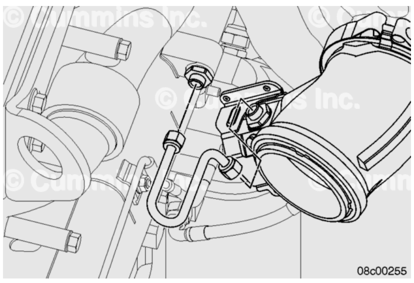
Remove

Automotive With CM871

Disconnect the aftertreatment injector coolant supply line at the inlet to the aftertreatment injector.

Disconnect the aftertreatment injector coolant supply line at the EGR cooler.

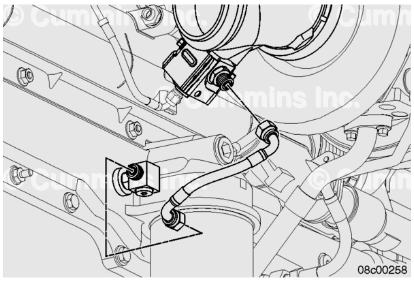
Remove the aftertreatment injector coolant supply line.

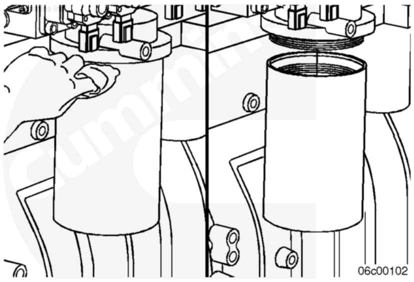


Disconnect the aftertreatment injector coolant return line at the aftertreatment injector.

Disconnect the aftertreatment injector coolant return line at the oil cooler.

Remove the aftertreatment injector coolant return line.





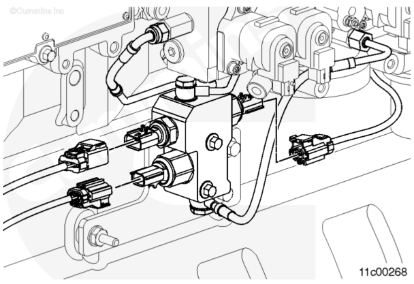
⚠ WARNING ⚠

Depending on the circumstance, diesel fuel is flammable. When inspecting or performing service or repairs in the fuel system, to reduce the impossibility of fire and resulting severe personal injury, death or property damage, never smoke or allow sparks or flames (such as pilot lights, electrical switches, or welding equipment) in the work area.

Clean the area around the fuel filter head and filter.

Remove the fuel filter with filter wrench, Part Number 3375249.

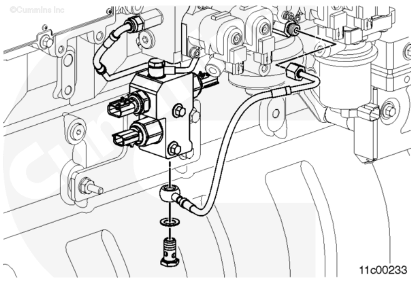
Disconnect the wiring harness from the shut off valve, fuel drain valve, and pressure sensor.



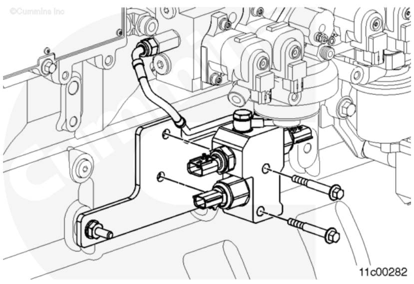
Disconnect the aftertreatment injector fuel supply line from the supply outlet of the fuel rifle.

Disconnect the aftertreatment injector fuel supply line at the inlet to the fuel shutoff manifold.

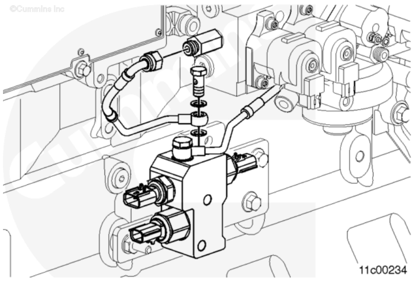
Remove the aftertreatment injector fuel supply line.



For access lower the fuel shutoff manifold by removing the two mounting capscrews.



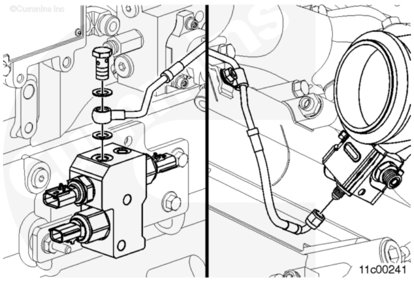
Disconnect the fuel return line from the fuel shutoff manifold and the fuel control module.



Disconnect the aftertreatment injector fuel supply line at the outlet of the fuel shutoff manifold.

Disconnect the fuel lines from the union fitting, located just above the EGR valve.

Disconnect the aftertreatment injector fuel supply line at the inlet to the aftertreatment injector, use a 9/16 inch wrench. In some applications, access to the aftertreatment injector is difficult and requires removal of the exhaust adapter pipe. Refer to Procedure 011-043 in Section 11. (/qs3/pubsys2/xml/en/procedures/101/101-011-043-tr.html)

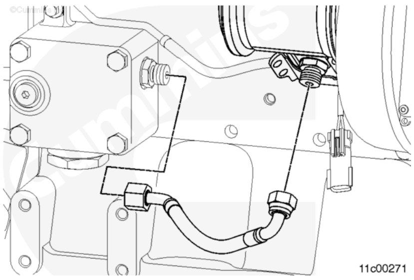


Automotive with CM876

Disconnect the aftertreatment injector coolant supply line at the inlet to the aftertreatment injector.

Disconnect the aftertreatment injector coolant supply line at the coolant heater housing.

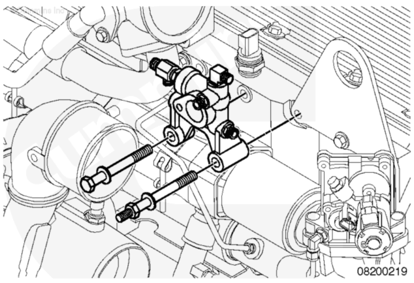
Remove the aftertreatment injector coolant supply line.



Disconnect the aftertreatment injector coolant return line at the aftertreatment injector.

Disconnect the aftertreatment injector coolant return line at the coolant return manifold.

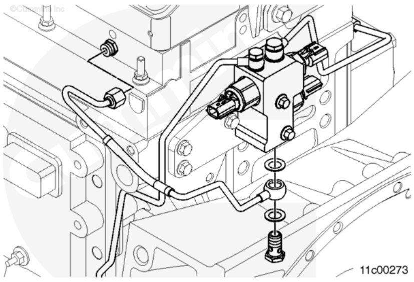
Remove the aftertreatment injector coolant return line.



Disconnect the aftertreatment injector fuel supply line from the supply outlet of the fuel rifle.

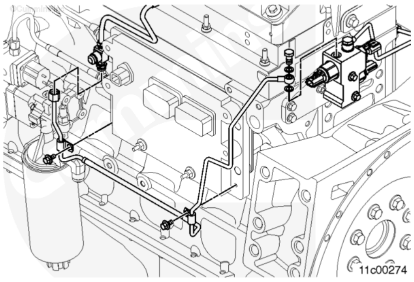
Disconnect the aftertreatment injector fuel supply line at the inlet to the fuel shutoff manifold.

Remove the aftertreatment injector fuel supply line.



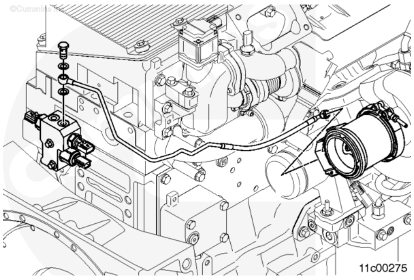
Disconnect the fuel return line from the fuel shutoff manifold and the fuel return tee fitting, located in front of the engine control module.

Remove the P-clips.



Disconnect the aftertreatment injector fuel supply line at the outlet of the fuel shutoff manifold.

Disconnect the aftertreatment injector fuel supply line at the inlet to the aftertreatment injector. In some applications, access to the aftertreatment injector is difficult and requires removal of the exhaust adapter pipe. Refer to Procedure 011-043 in Section 11. (/qs3/pubsys2/xml/en/procedures/101/101-011-043-tr.html)



Clean and Inspect for Reuse

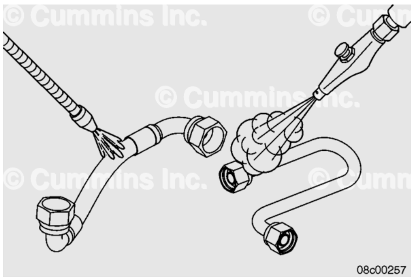
Automotive With CM871

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacture's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

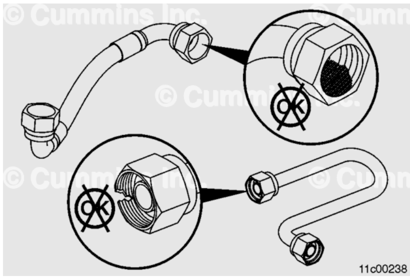


Clean the aftertreatment injector coolant lines with a safety solvent.

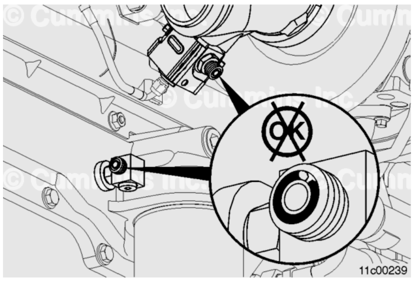
Inspect the aftertreatment injector coolant line threads for damage. Replace if necessary.

Make sure the aftertreatment injector coolant lines are clear of debris.

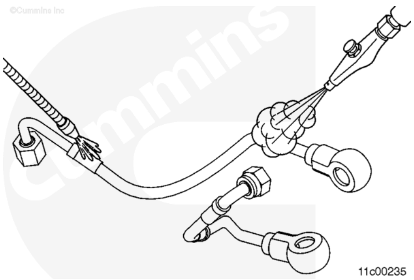
Use compressed air to flush the lines and remove any loose dirt particles.



Check the o-ring for cuts and wear. Replace if necessary.



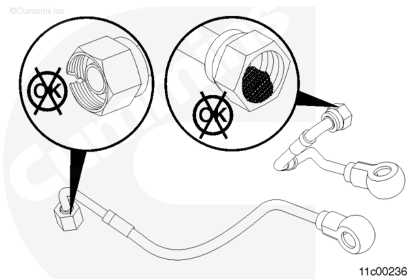
Clean the aftertreatment injector fuel lines with a safety solvent.



Inspect the threads and banjo fittings on the aftertreatment injector fuel lines. Replace is necessary.

Make sure that the aftertreatment injector fuel lines are clear of debris.

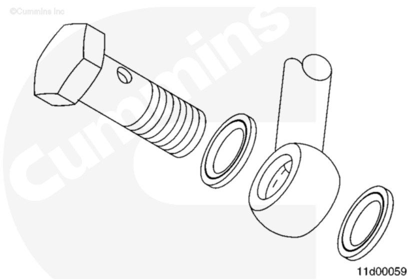
Use compressed air to flush the lines and remove any loose dirt particles.



Inspect the face of the banjo connection for sealing surface damage.

Inspect the straight thread fittings and fuels lines for cracks, bends or other damage.

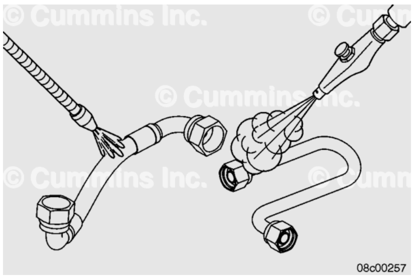
Replace as necessary.



Automotive with CM876

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacture's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

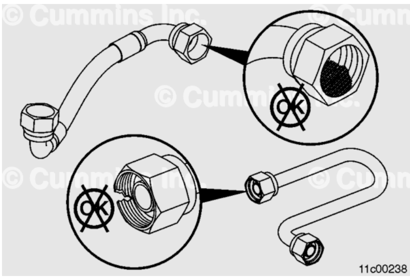
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Clean the aftertreatment injector coolant lines with a safety solvent.

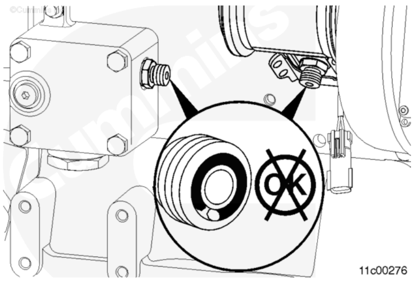
Inspect the aftertreatment injector coolant line threads for damage. Replace if necessary.

Make sure the aftertreatment injector coolant lines are clear of debris.

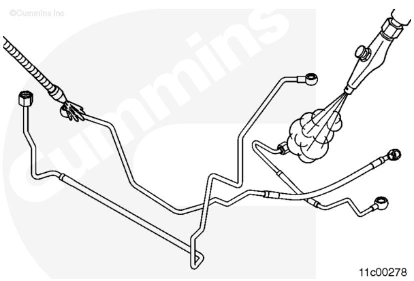
Use compressed air to flush the lines and remove any loose dirt particles.



Check the o-ring for cuts and wear. Replace if necessary.



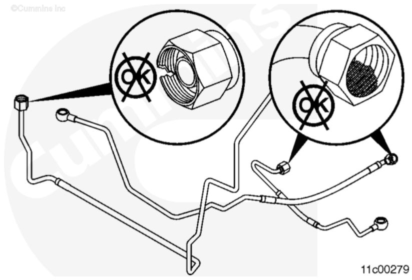
Clean the aftertreatment injector fuel lines with a safety solvent.



Inspect the threads and banjo fittings on the aftertreatment injector fuel lines. Replace if necessary.

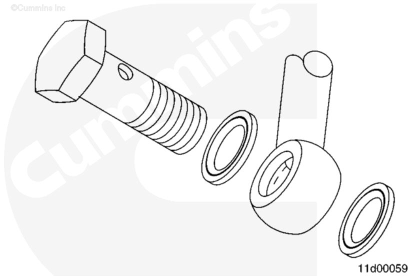
Make sure the aftertreatment injector fuel lines are clear of debris.

Use compressed air to flush the lines and remove any loose dirt particles.



Inspect the face of the banjo connector for sealing surface damage.

Inspect the straight thread fittings and fuels for cracks, bends or other damage.

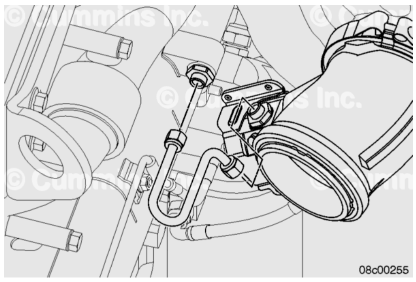


Install

Automotive With CM871

Connect the aftertreatment injector coolant supply line to the aftertreatment injector and the EGR cooler.

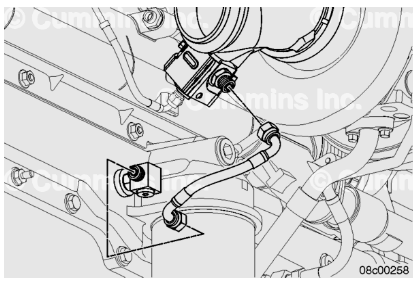
Torque Value: 24 n•m [212 in-lb]



Connect the aftertreatment injector coolant return line to the oil cooler.

Connect the aftertreatment injector coolant return line to the aftertreatment injector.

Torque Value: 24 n•m [212 in-lb]



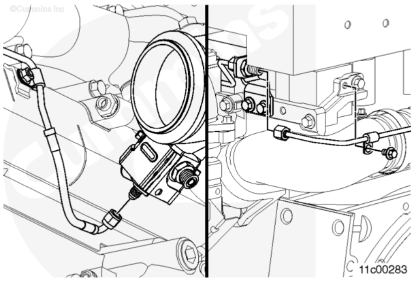
Connect the aftertreatment injector fuel supply line to the aftertreatment injector.

Torque Value: 24 n•m [212 in-lb]

Connect the aftertreatment injector fuel supply line to the outlet of the union fitting, located just above the EGR valve.

Torque Value: 24 n•m [212 in-lb]

Install the P-clips.



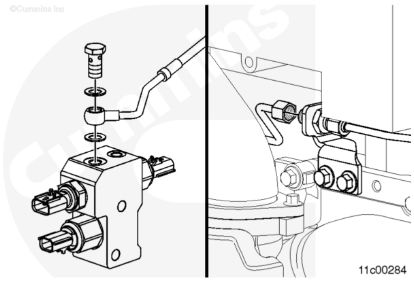
Connect the aftertreatment injector fuel supply line to the inlet of the union fitting, located just above the EGR valve.

Torque Value: 24 n•m [212 in-lb]

Connect the aftertreatment injector fuel supply line to the outlet of the fuel shutoff manifold.

Torque Value: 24 n•m [212 in-lb]

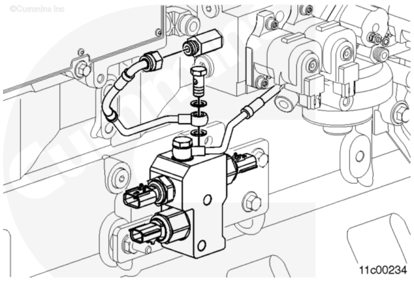
Install the P-clips.



Connect the fuel return line to the fuel shutoff valve manifold and the fuel control module.

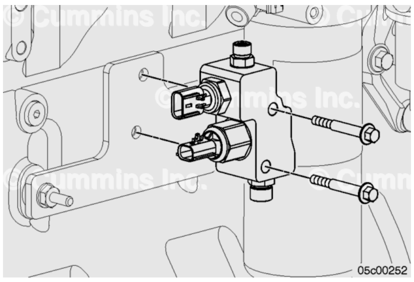
Torque Value:
O-ring joint 24 n•m [212 in-lb]

Torque Value:
Banjo joint 15 n•m [133 in-lb]



Install the two capscrews to attach the aftertreatment fuel shutoff manifold to the bracket on the engine.

Torque Value: 24 n•m [212 in-lb]

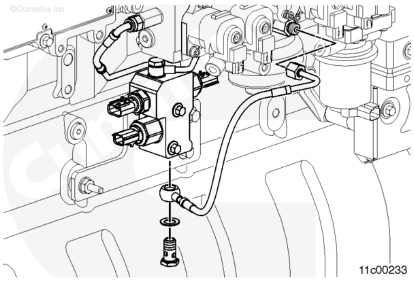


Connect the aftertreatment injector fuel supply line to the inlet of the shutoff manifold.

Torque Value:
Banjo joint 34 n•m [25 ft-lb]

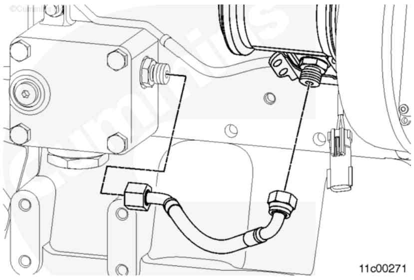
Connect the aftertreatment injector fuel supply line to the fuel rifle.

Torque Value:
O-ring joint 24 n•m [212 in-lb]



Automotive with CM876

Connect the aftertreatment injector coolant supply line to the aftertreatment injector and the coolant heater housing.

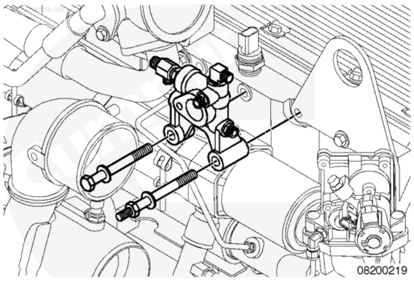


Connect the aftertreatment injector coolant return line to the coolant return line to the coolant return manifold.

Connect the aftertreatment injector coolant return line to the aftertreatment injector.

Install the P-clips.

Torque Value: 24 n•m [212 in-lb]

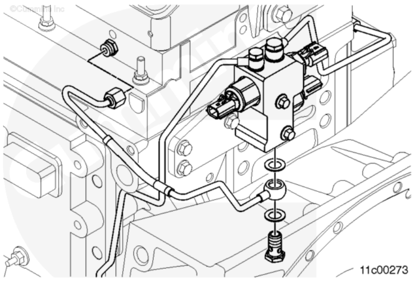


Connect the aftertreatment injector fuel supply line to the aftertreatment injector.

Torque Value: 24 n•m [212 in-lb]

Connect the aftertreatment injector fuel supply line to the outlet of the fuel shutoff manifold.

Torque Value: 34 n•m [25 ft-lb]

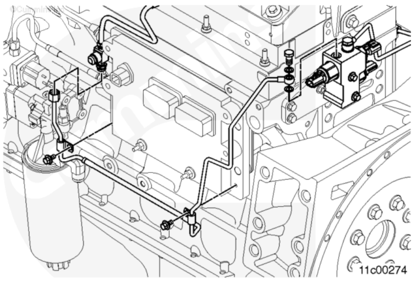


Connect the fuel return line to the fuel shutoff valve manifold and the fuel return tee fitting, located in front of the engine control module.

Install the P-clips

Torque Value:
O-ring fitting 24 n•m [212 in-lb]

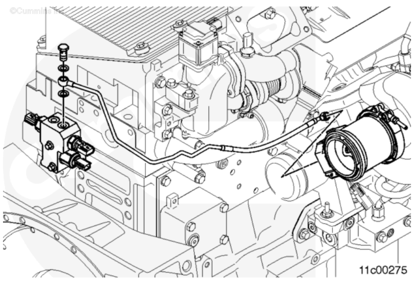
Torque Value:
Banjo fitting 15 n•m [133 in-lb]



Connect the aftertreatment injector fuel supply line to the inlet of the shutoff valve manifold.

Torque Value: 24 n•m [212 in-lb]

Connect the aftertreatment injector fuel supply line to the fuel rifle.



Torque Value: 24 n•m [212 in-lb]

Finishing Steps

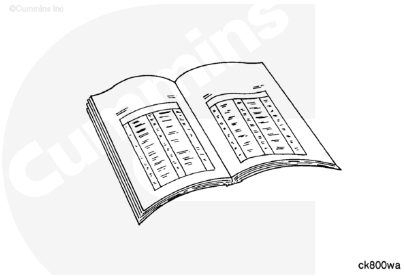
Automotive With CM871

Connect the wiring harness to pressure sensor, shutoff valve, and fuel drain valve.

Install the fuel filter.

Fill the cooling system. Refer to Procedure 008-018 (/qs3/pubsys2/xml/en/procedures/10/10-008-018-tr.html).

Operate the engine and check for leaks.



Automotive with CM876

Fill the cooling system. Refer to Procedure 008-018 (/qs3/pubsys2/xml/en/procedures/35/35-008-018-tr.html).

Operate the engine and check for leaks.

Last Modified: 28-Mar-2012